Project Title: Multi modal portable agricultural garden

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Abstract: Modern agriculture is composed of agricultural innovations and farming practices that increase efficiency with minimum dependence and exploitation of natural resources. We are a group of students involved in experimenting with modern agricultural practices, based on personal experience and training.

The major aim of our plan and proposal is to create a multidimensional and portable vegetable garden, which is a closed vertical system. We believe cycling metabolites between closed systems of farming helps create self-sustaining units with minimum waste, less energy expenditure and better yield. The main objective of the study is to combine fish culture with vegetable cultivation, which supply majority of greens in a household along with recreation. For the vertical garden, the GI frame is employed. A water tank is set for the aquarium. A stair or pyramid-like model is provided for planting vegetables. Portability is ensured with placement of wheels. Electrical and plumbing units are arranged to ensure circulation of water and nutrients. There is minimal utilization of water, energy and feed. The system allows cycling of aquaculture waste, a source of nutrition for the vegetables grown in the unit. Water enriched with soil after watering the plants is to be resurfaced into the aquarium.

Thus our plan is recreational, energy efficient, flexible and marketable.